Applied Machine Learning for Industry Solutions

UNIT 2 - 5 MARK QUESTION AND ANSWER BANK

Prepared from the uploaded Unit 2 study material and Unit 2 question bank.

Exam-oriented answers written in simple language for 5-mark questions.

Note: *The answers are kept source-based. Extra explanation is added only where necessary to connect the steps of the given question.*

Q1. Apply PCA to the Iris dataset to reduce features to 2 principal components and visualize the results.

5-Mark Answer

Answer: Principal Component Analysis (PCA) is used to reduce the number of features while preserving most of the important information in the dataset.

For the Iris dataset, PCA is applied to convert the original 4 feature dimensions into 2 principal components.

Steps:

• Load the Iris dataset.

• Separate the input features and target variable.

• Apply PCA with **n_components = 2**.

• Transform the original data into two principal components.

• Create a 2D scatter plot using Principal Component 1 and Principal Component 2.

• Use the target/species values to distinguish the classes in the plot.

Source code pattern:

```
from sklearn.datasets import load_iris
from sklearn.decomposition import PCA
import pandas as pd
import matplotlib.pyplot as plt

iris = load_iris()
X = iris.data
y = iris.target

pca = PCA(n_components=2)
principalComponents = pca.fit_transform(X)

principalDf = pd.DataFrame(
    principalComponents,
    columns=['principal component 1',
             'principal component 2']
)

plt.scatter(
    principalDf['principal component 1'],
    principalDf['principal component 2'],
    c=y, cmap='viridis', edgecolor='k', s=70
)
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("Iris Data in 2D after Applying PCA")
plt.show()
```

Interpretation: After PCA, the 4-dimensional Iris data is visualized in 2D. Principal Component 1 contains about **62.4%** of the variance and Principal Component 2 contains about **34.0%**. Together, the two components preserve about **96.40%** of the information.

Q2. Explain the Exploratory Data Analysis (EDA) and its Techniques with examples.

5-Mark Answer

Answer: Exploratory Data Analysis (EDA) is the process of summarizing and visualizing a dataset to understand its main characteristics before building a machine learning model.

Goals of EDA:

• Understand data structure and distribution.

• Identify missing values, outliers, and anomalies.

• Detect relationships between features.

• Test basic hypotheses.

• Choose suitable preprocessing steps.

Main EDA techniques:

1. Data overview: Use functions such as **head()**, **info()**, **describe()**, and missing-value checks to understand the dataset.

2. Univariate analysis: Study one variable at a time. A histogram can show the distribution of a numerical feature such as Age. A boxplot can help detect outliers. A countplot or value counts can be used for categorical features.

3. Bivariate analysis: Study the relationship between two variables. A scatterplot can be used for numerical variables such as Age and Fare.

4. Correlation analysis: A correlation matrix and heatmap show the relationship between numerical variables.

5. Hypothesis testing: A statistical test can be used to check whether there is enough evidence for a stated assumption.

Example: The Unit 2 material uses the Titanic dataset. A histogram of Age can show the age distribution, a boxplot of Fare can show possible outliers, a scatterplot of Age versus Fare can show patterns, and a correlation matrix can show relationships between numerical variables.

Q3. Build a Decision Tree for the cosmetics-shop dataset with Buys as the target variable. Find the root node and give the decision for the test data [Age < 21, Income = Low, Gender = Female, Marital Status = Married].

5-Mark Answer

Answer: A Decision Tree is a supervised learning model that uses a tree-like structure to make decisions. The best root node is selected using the attribute with the highest Information Gain.

Step 1: Calculate the entropy of the complete dataset.

The dataset contains 14 records: **9 Yes** and **5 No**. Therefore:

Entropy(S) = -(9/14) log2(9/14) - (5/14) log2(5/14) = approximately 0.94

Step 2: Calculate Information Gain for each attribute.

• Gain(Age) = 0.247

• Gain(Income) = 0.029

• Gain(Gender) = 0.024

• Gain(Marital Status) = 0.048

Step 3: Select the root node. Age has the highest Information Gain, so the **root node is Age**.

Step 4: Decision for the given test data. The test record is **Age < 21, Income = Low, Gender = Female, Marital Status = Married**. In the training data, the same combination appears with the decision **Buys = Yes**.

Final Decision: Buys = Yes.

Conclusion: The Decision Tree selects **Age** as the root because it has the highest Information Gain, and the given customer is predicted to **buy the product**.

Q4. Apply PCA on the MNIST dataset to reduce its dimensionality, visualize the data in 2D, and analyze the impact on model training time and accuracy.

5-Mark Answer

Answer: PCA can be used to reduce the dimensionality of the MNIST handwritten-digit dataset before applying a machine learning algorithm.

The MNIST dataset is suitable because it has **784 feature columns**, a training set of **60,000 examples**, and a test set of **10,000 examples**.

Steps:

• Load the MNIST training and test data.

• Separate the input features and digit labels.

• Apply PCA to reduce the original 784-dimensional data to fewer principal components.

• For 2D visualization, transform the selected data into 2 principal components and plot Principal Component 1 versus Principal Component 2.

• Train Logistic Regression on the original data and on the PCA-reduced data.

• Compare the training time and accuracy.

2D visualization pattern:

```
pca = PCA(n_components=2)
X_2d = pca.fit_transform(X)

plt.scatter(X_2d[:, 0], X_2d[:, 1], c=y)
plt.xlabel("Principal Component 1")
plt.ylabel("Principal Component 2")
plt.title("MNIST Data in 2D after PCA")
plt.show()
```

Impact: PCA reduces the number of dimensions, so the machine learning algorithm has fewer input features to process. Therefore, the main purpose is to **speed up model training**. Because dimensionality reduction removes some information, the accuracy can change depending on how much variance is retained.

Conclusion: PCA is useful for high-dimensional datasets such as MNIST because it reduces computational complexity and can make training faster while retaining the most important variation in the data.

Q5. Implement Gradient Boosting on a real-world dataset and illustrate how boosting iteratively improves weak learners to build a strong predictive model.

5-Mark Answer

Answer: Gradient Boosting Machine (GBM) is an ensemble learning technique used for classification, regression, and ranking tasks. It builds weak prediction models, usually shallow Decision Trees, sequentially.

Real-world dataset: The Unit 2 material demonstrates Gradient Boosting using a customer-churn dataset with 7043 rows and 21 columns.

Working of boosting:

• Start with a weak model, usually a shallow Decision Tree.

• Compute residuals, which are the errors between the true and predicted values.

• Train the next weak model on these residuals so that it corrects previous errors.

• Add the new model to the ensemble using a weight controlled by the learning rate.

• Repeat the process for a fixed number of iterations or until convergence.

Implementation pattern:

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.30, random_state=42
)

gb_model = GradientBoostingClassifier(
    n_estimators=100,
    learning_rate=0.1,
    max_depth=3,
    random_state=42
)

gb_model.fit(X_train, y_train)
y_pred = gb_model.predict(X_test)
```

Interpretation: The model does not build all trees independently. Each new tree tries to improve the errors of the previous model. This sequential improvement converts many weak learners into a stronger predictive model.

The Unit 2 example reports an accuracy of approximately **0.80047**.

Q6. Using the given ages of website visitors, apply K-Means clustering to group them into suitable clusters and interpret the results.

5-Mark Answer

Answer: Given ages:

[16, 16, 17, 20, 20, 21, 21, 22, 23, 29, 36, 41, 42, 43, 44, 45, 61, 62, 66]

Choose **K = 2**. The Unit 2 example initially selects two random centroids: **C1 = 16** and **C2 = 22**.

Iteration 1: C1 = 16.33 for [16, 16, 17]; C2 = 37.25 for [20, 20, 21, 21, 22, 23, 29, 36, 41, 42, 43, 44, 45, 61, 62, 66].

Iteration 2: C1 = 19.55 for [16, 16, 17, 20, 20, 21, 21, 22, 23]; C2 = 46.90 for [29, 36, 41, 42, 43, 44, 45, 61, 62, 66].

Iteration 3: C1 = 20.50 for [16, 16, 17, 20, 20, 21, 21, 22, 23, 29]; C2 = 48.89 for [36, 41, 42, 43, 44, 45, 61, 62, 66].

Iteration 4: The centroids and members remain unchanged, so the algorithm stops.

Final clusters:

• Cluster 1: Ages 16 to 29, centroid = 20.50

• Cluster 2: Ages 36 to 66, centroid = 48.89

Interpretation: The website visitors are divided into two age-based groups. The first group contains younger visitors and the second group contains older visitors. The algorithm stops because there is no change between Iteration 3 and Iteration 4.

Q7. Apply Decision Trees on suitable industrial datasets and demonstrate their applications by interpreting the results for real-world decision-making.

5-Mark Answer

Answer: A Decision Tree is a supervised learning algorithm that uses a flowchart-like tree structure to make decisions. Internal nodes represent conditions, branches represent the result of a condition, and end nodes represent the final result.

Application procedure:

• Select a suitable industrial dataset.

• Identify input features and the target variable.

• Train the Decision Tree on the available data.

• The tree splits the data using conditions that help in prediction.

• Interpret the final branches and predicted result for decision-making.

Industrial applications from the Unit 2 material:

• **Healthcare:** Predicting patient readmission, disease diagnosis, and identifying genetic markers.

• **Finance:** Credit scoring, option pricing, risk assessment, and fraud detection.

• **Marketing:** Customer segmentation, churn prediction, and targeted marketing campaigns.

• **Business Strategy:** Visualization of decision processes and cost-effectiveness analyses.

Interpretation: The path followed from the root node to a leaf node shows the conditions that lead to the final prediction. Therefore, Decision Trees are useful when an industry needs a prediction and also a clear explanation of the decision process.

Q8. Implement K-Means, Hierarchical Clustering, and DBSCAN on a suitable dataset, compare their clustering results, and interpret the differences in terms of cluster formation and outlier detection.

5-Mark Answer

Answer: The three algorithms can be applied to the same standardized dataset and their results can be compared.

1. K-Means: Choose the number of clusters K. The algorithm assigns points to the nearest centroid and repeatedly recalculates centroids until there is no significant change.

2. Hierarchical Clustering: In the agglomerative method, each point starts as its own cluster. The closest clusters are merged step by step and the hierarchy is shown using a dendrogram.

3. DBSCAN: DBSCAN groups closely packed points using density. It uses two main parameters: **eps** and **min_samples**. Points marked with label **-1** are noise points.

Comparison of cluster formation:

• K-Means is centroid-based and requires the value of K.

• Hierarchical Clustering is tree-based and forms nested clusters.

• DBSCAN is density-based and can identify arbitrary-shaped clusters.

Comparison of outlier detection: DBSCAN automatically identifies noise or outlier points. K-Means and standard Hierarchical Clustering do not inherently mark noise in the same way.

Conclusion: K-Means is useful for centroid-based grouping, Hierarchical Clustering is useful for understanding nested relationships, and DBSCAN is useful when the dataset contains arbitrary-shaped clusters and noise.

Q9. Use histogram and scatterplot visualizations on a dataset to detect patterns, trends, and anomalies, and explain the insights obtained.

5-Mark Answer

Answer: Histograms and scatterplots are important EDA visualization techniques. The Unit 2 material uses the Titanic dataset for examples.

1. Histogram: A histogram is used to understand the distribution of a numerical variable.

```
df['age'].hist(bins=30, edgecolor='black')
plt.title('Age Distribution')
plt.xlabel('Age')
plt.ylabel('Frequency')
plt.show()
```

Insight from histogram: It can show the distribution of Age, the frequency of different age ranges, and possible skewness or unusual gaps in the data.

2. Scatterplot: A scatterplot is used to study the relationship between two numerical variables.

```
sns.scatterplot(x='age', y='fare', data=df)
plt.title('Age vs Fare')
plt.show()
```

Insight from scatterplot: It can show whether Age and Fare have a visible relationship, whether there are groups or trends, and whether some points appear unusually far from the main pattern.

Conclusion: Histogram is mainly used to understand the distribution of one numerical feature, while scatterplot is used to understand the relationship between two numerical features and to identify patterns and anomalies.

Q10. Apply correlation analysis on a dataset to identify relationships between variables, and interpret the strength and direction of these relationships.

5-Mark Answer

Answer: Correlation analysis is used to study the relationship between numerical variables. A correlation matrix can be calculated and visualized using a heatmap.

```
correlation = df.corr(numeric_only=True)

sns.heatmap(
    correlation,
    annot=True,
    cmap='coolwarm'
)
plt.title('Correlation Matrix')
plt.show()
```

Interpretation of correlation values:

• **+1:** The two variables move upward together. This indicates a positive relationship.

• **-1:** When one variable increases, the other decreases. This indicates a negative relationship.

• **0:** There is no linear relationship.

Strength: A correlation value closer to +1 or -1 indicates a stronger relationship, while a value closer to 0 indicates a weak or no linear relationship.

Conclusion: Correlation analysis helps identify the direction and strength of relationships between numerical variables. The heatmap makes these relationships easier to interpret.

Q11. Conduct hypothesis testing on a relevant dataset to validate an assumption, and interpret the statistical significance of the results.

5-Mark Answer

Answer: Hypothesis testing is a statistical method used to decide whether there is enough evidence to infer that a certain condition is true for a population.

Example: Test whether the mean fare paid by male and female passengers is significantly different.

Null hypothesis (H0): The mean fare paid by males and females is the same.

Alternative hypothesis (H1): The mean fare paid by males and females is different.

```
from scipy.stats import ttest_ind

male_fare = df[df['sex'] == 'male']['fare'].dropna()
female_fare = df[df['sex'] == 'female']['fare'].dropna()

t_stat, p_val = ttest_ind(male_fare, female_fare)

print(f"T-statistic: {t_stat}, P-value: {p_val}")
```

Output in the Unit 2 material:

• T-statistic = -5.529140269385719

• P-value = 4.2308678700429995e-08

Interpretation rule:

• If p-value < 0.05, reject H0. There is a statistically significant difference.

• If p-value is greater than or equal to 0.05, fail to reject H0. There is no strong evidence of a difference.

Since the given p-value is much smaller than 0.05, **H0 is rejected**. Therefore, there is a statistically significant difference in the mean fare paid by male and female passengers.

Q12. Apply Single Linkage, Complete Linkage, and Average Linkage hierarchical clustering methods on a sample dataset. Generate dendrograms for each method and compare the clustering structures.

5-Mark Answer

Answer: In Hierarchical Clustering, linkage decides how the distance between two clusters is calculated. The clustering structure can be visualized using a dendrogram.

1. Single Linkage: The distance between two clusters is the **shortest distance** between a point in one cluster and a point in the other cluster.

2. Complete Linkage: The distance between two clusters is the **longest distance** between a point in one cluster and a point in the other cluster.

3. Average Linkage: The distance between two clusters is the **average distance** between every point in one cluster and every point in the other cluster.

Implementation pattern:

```
from scipy.cluster.hierarchy import linkage, dendrogram
import matplotlib.pyplot as plt

linked_single = linkage(X_scaled, method='single')
linked_complete = linkage(X_scaled, method='complete')
linked_average = linkage(X_scaled, method='average')

dendrogram(linked_single)
plt.title("Single Linkage Dendrogram")
plt.show()

dendrogram(linked_complete)
plt.title("Complete Linkage Dendrogram")
plt.show()

dendrogram(linked_average)
plt.title("Average Linkage Dendrogram")
plt.show()
```

Comparison: Single Linkage joins clusters based on the nearest points, Complete Linkage uses the farthest points, and Average Linkage uses the average distance. Therefore, the merging order and the final dendrogram structure can differ for the same dataset.

Conclusion: By comparing the three dendrograms, we can understand how the choice of linkage changes the hierarchical cluster structure.

Q13. Apply Random Forest on suitable industrial datasets, and demonstrate its applications by interpreting the results for real-world decision-making.

5-Mark Answer

Answer: Random Forest is an ensemble learning method that combines multiple Decision Trees to improve predictive accuracy and control overfitting. It is used for classification and regression.

Working:

• Create many Decision Trees.

• Train each tree using a random sample of the training data through Bootstrap Aggregating (Bagging).

• During splitting, consider a random subset of features.

• For classification, combine predictions using majority voting.

• For regression, calculate the average prediction.

Industrial applications:

• **Insurance:** Risk assessment of customers and claims.

• **Banking/Finance:** Credit scoring, fraud detection, loan default prediction, and stock forecasting.

• **Healthcare:** Disease prediction using genetic and imaging data.

• **Marketing:** Customer segmentation, predictive marketing, and recommendation systems.

• **Text/Image Classification:** Spam detection, document or message categorization, and image recognition.

Interpretation: A trained Random Forest can predict the class or numerical value for new industrial data. Feature importance can also show which features are more useful for prediction.

Conclusion: Random Forest is suitable for real-world decision-making because it combines the predictions of many trees and is more resistant to overfitting than a single Decision Tree.

Q14. Apply Gradient Boosting Machines on suitable industrial datasets, and demonstrate their applications by interpreting the results for real-world decision-making.

5-Mark Answer

Answer: Gradient Boosting Machines (GBMs) are ensemble learning models used for classification, regression, and ranking. They build weak models, usually Decision Trees, sequentially.

Application procedure:

• Select an industrial dataset and identify the features and target variable.

• Split the data into training and testing sets.

• Start with a weak Decision Tree.

• Compute residual errors.

• Train the next tree to correct these errors.

• Repeat the process and combine the models using the learning rate.

Unit 2 implementation pattern:

```
gb_model = GradientBoostingClassifier(
    n_estimators=100,
    learning_rate=0.1,
    max_depth=3,
    random_state=42
)

gb_model.fit(X_train, y_train)
y_pred = gb_model.predict(X_test)
```

Interpretation: The model improves prediction step by step because each new tree focuses on the errors made by the previous model. Feature-importance values can be used to identify important input features.

Decision-making: The Unit 2 material demonstrates GBM on a real-world customer-churn dataset. The predicted output can help identify likely outcomes from customer information, and the feature-importance values help interpret which inputs are important.

Conclusion: GBM is useful when high predictive accuracy is important, especially for structured or tabular industrial datasets, but it is slower to train and requires careful parameter tuning.

Q15. Apply different distance metrics (Euclidean, Manhattan, Minkowski) in a clustering algorithm and analyze how each metric influences the resulting cluster formation.

5-Mark Answer

Answer: Distance measures are used in clustering to calculate similarity or dissimilarity between data points. A lower distance generally indicates that points are more similar.

1. Euclidean Distance: This is the straight-line distance between two points.

For two-dimensional points (x1, y1) and (x2, y2): **d = sqrt[(x2 - x1)^2 + (y2 - y1)^2]**

2. Manhattan Distance: This measures the total absolute difference along each dimension.

d = |x2 - x1| + |y2 - y1|

3. Minkowski Distance: This is a generalized distance measure.

d = [sum |xi - yi|^p]^(1/p)

Influence on clustering:

• The selected distance metric changes how similarity between points is measured.

• Because the calculated distances can change, the nearest points or nearest clusters can also change.

• Therefore, the resulting cluster formation can differ when Euclidean, Manhattan, or Minkowski distance is used.

Conclusion: The choice of distance metric is important because clustering algorithms use distance to decide which data points are similar and should belong to the same cluster.

Q16. Apply K-Means, Hierarchical Clustering, and DBSCAN on suitable industrial datasets, and analyze their clustering results in the context of real-world applications.

5-Mark Answer

Answer: The three clustering techniques can be selected according to the structure and requirement of industrial data.

K-Means: It is centroid-based and divides data into K groups. It is suitable for customer segmentation, retail product clustering, social-media user grouping, bioinformatics, document clustering, and speech recognition.

Hierarchical Clustering: It creates nested clusters and a dendrogram. It is useful when hierarchical relationships are important. Applications include customer segmentation, healthcare patient subgroups, finance, telecommunications, and recruitment.

DBSCAN: It is density-based and can identify closely packed clusters and noise. Applications include geospatial hotspot detection, financial fraud detection, manufacturing/IoT anomaly detection, healthcare sensor analysis, and image segmentation.

Analysis:

• Use K-Means when a fixed number of centroid-based groups is required.

• Use Hierarchical Clustering when a hierarchy of relationships between groups is useful.

• Use DBSCAN when the data may contain arbitrary-shaped clusters and outliers.

Conclusion: The industrial context decides which clustering technique is more suitable because each method forms clusters differently and handles noise differently.

Q17. Implement a Random Forest classifier on a relevant dataset to perform classification, and interpret how the model makes predictions.

5-Mark Answer

Answer: A Random Forest classifier is created by combining multiple Decision Trees.

Implementation pattern from Unit 2:

```
# Features and target
X = df.iloc[:, :-1]
y = df.iloc[:, -1].values

# Encode categorical columns
X = X.apply(LabelEncoder().fit_transform)

# Create Random Forest Classifier
rf_model = RandomForestClassifier(
    n_estimators=10,
    criterion='entropy',
    random_state=0
)

rf_model.fit(X.iloc[:, 1:5], y)

# Predict a new value
X_in = np.array([0, 1, 0, 1])
y_pred = rf_model.predict([X_in])
```

How the model makes predictions:

• Many Decision Trees are created.

• Each tree is trained on a random sample of the training data using Bagging.

• A random subset of features is considered while splitting nodes.

• For a new input, every tree gives its own prediction.

• The final classification is obtained by **majority voting**.

Feature importance: The Unit 2 example gives the following values: Age = 0.4413, Income = 0.1263, Gender = 0.2530, and Marital Status = 0.1795.

Interpretation: In this example, Age has the highest importance. The final class prediction is based on the combined voting of the trees.

Q18. Compare the performance of Gradient Boosting and Random Forest on the same dataset, and analyze differences in accuracy, feature importance, and interpretability.

5-Mark Answer

Answer: Random Forest and Gradient Boosting are both ensemble methods based on Decision Trees, but they build and combine trees differently.

Random Forest: Uses Bagging. It builds many Decision Trees using random samples and random feature selection. For classification, it uses majority voting.

Gradient Boosting: Builds trees sequentially. Each new tree focuses on the errors of the previous model.

Accuracy: Compare the accuracy scores of both models on the same test data. The Unit 2 material states that Gradient Boosting is often more accurate than Random Forest for structured or tabular data, but the actual result should be taken from the applied dataset.

Feature importance: Both models can provide feature-importance values. Compare these values to identify which features each model considers important for prediction.

Interpretability: Both are less directly interpretable than a single Decision Tree. Random Forest combines many trees, while Gradient Boosting creates a sequential ensemble, making both more complex than one tree.

Training speed: Random Forest is generally faster to train. Gradient Boosting is slower because its trees are trained sequentially.

Conclusion: Random Forest is useful for robust performance and controlling overfitting, while Gradient Boosting is useful when high accuracy is more important and careful tuning is possible.

Q19. Apply K-Means clustering to group customers of a grocery store based on purchase frequency and basket size, and visualize the clusters.

5-Mark Answer

Answer: K-Means can group grocery-store customers using two features: **purchase frequency** and **basket size**. The purpose is to place similar customers into the same cluster.

Steps:

• Create a dataset containing purchase frequency and basket size.

• Select the number of clusters K.

• Apply K-Means to the two customer features.

• Assign each customer to the nearest centroid.

• Recalculate centroids and repeat until the clusters become stable.

• Visualize the customers using a scatter plot, with cluster labels shown by different groups and centroids shown separately.

Implementation pattern:

```
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

X = df[['purchase_frequency', 'basket_size']]

kmeans = KMeans(
    n_clusters=3,
    random_state=42
)

labels = kmeans.fit_predict(X)

plt.scatter(
    X['purchase_frequency'],
    X['basket_size'],
    c=labels
)

centers = kmeans.cluster_centers_
plt.scatter(
    centers[:, 0],
    centers[:, 1],
    marker='X',
    s=100
)

plt.xlabel("Purchase Frequency")
plt.ylabel("Basket Size")
plt.title("Grocery Store Customer Clusters")
plt.show()
```

Interpretation: Customers placed in the same cluster have similar purchase frequency and basket size. The clusters can therefore be used to understand different customer groups for customer segmentation and targeted marketing.

Q20. Apply the DBSCAN clustering algorithm on a suitable dataset to identify clusters and outliers, and interpret the results.

5-Mark Answer

Answer: DBSCAN stands for **Density-Based Spatial Clustering of Applications with Noise**. It groups closely packed data points and marks outliers as noise.

Main parameters:

• **eps:** Radius used to check nearby points.

• **min_samples:** Minimum density required for a core point.

Implementation pattern from Unit 2:

```
from sklearn.cluster import DBSCAN

dbscan = DBSCAN(
    eps=0.5,
    min_samples=5
)

labels_dbscan = dbscan.fit_predict(X_scaled)

plt.scatter(
    X_scaled[:, 0],
    X_scaled[:, 1],
    c=labels_dbscan,
    cmap='tab10',
    s=10
)

plt.title("DBSCAN Clustering")
plt.show()

n_noise = np.sum(labels_dbscan == -1)
print(
    f"Number of noise points detected by DBSCAN: {n_noise}"
)
```

Interpretation:

• Data points in dense regions are grouped into clusters.

• DBSCAN does not assume spherical clusters, so it can identify arbitrary-shaped clusters.

• Points labelled **-1** are treated as noise or outliers.

The Unit 2 example detects **75 noise points**.

Conclusion: DBSCAN is useful when a dataset contains dense groups together with isolated points or anomalies.

Q21. Implement K-Means, Hierarchical Clustering, and DBSCAN on a suitable dataset, compare their clustering results, and interpret the differences in terms of cluster formation and outlier detection.

5-Mark Answer

Answer: This question is the same comparison as Q8. The three methods can be applied to the same standardized dataset and the results can be compared.

K-Means:

• Centroid-based method.

• Requires the number of clusters K.

• Each point is assigned to the nearest centroid.

• Centroids are recalculated until the algorithm stabilizes.

Hierarchical Clustering:

• Tree-based method.

• Agglomerative clustering starts with each point as a separate cluster and merges the closest clusters.

• The result can be visualized using a dendrogram.

DBSCAN:

• Density-based method.

• Uses eps and min_samples.

• Groups closely packed points and identifies isolated points as noise.

Comparison: K-Means forms clusters around centroids, Hierarchical Clustering forms nested clusters based on successive merging or splitting, and DBSCAN forms clusters based on density.

Outlier detection: DBSCAN directly identifies noise points using the label **-1**. K-Means and standard Hierarchical Clustering do not inherently identify noise in the same way.

Conclusion: K-Means is useful for centroid-based grouping, Hierarchical Clustering is useful for studying hierarchical relationships, and DBSCAN is useful when outlier detection and arbitrary cluster shapes are important.